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*Assignment: Study of the change in the solar
cell performance when there is a change in
tilt angle*

Introduction and Methodology: PVsyst is widely used in the field of renewable energy for optimal designing of solar power plants and to estimate the energy yielded.

In this assignment, there are a total of 57 optimizers which are equally distributed among three strings (19 optimizers in each) of the solar panel which is in turn attached to the inverter 9. The Eelde weather data was first downloaded. A period between 1st October 2022 to 31st September 2023 was considered. The weather sheet was imported 20 as the tilt angle was entered and the system as well as the detailed losses were checked. The relative daily difference along with average value was calculated. Again, the tilt angle was changed by 10% (in this assignment a 10% decrease is executed), and then again another excel sheet was obtained after running the stimulation. At last, both the obtained data are compared in order to observe change that happened due to the change in the orientation.

Results:

The average relative daily difference for the optimizer 9.1.3 is approximately -2.5079 and the average relative daily difference for the optimizer 9.1.14 is approximately -3.273. This value is obtained by taking the average of all the relative daily differences in energy yield (for both the optimizers). The relative daily difference was calculated by subtracting the actual value from Entrance from the stimulated value and then dividing it with the stimulated value (for the percentage a 100 was multiplied to it). Before doing so the stimulated value needs to be divided by the total number of optimizers so that the value per optimizer can be achieved.

The highest value of the relative daily difference is somewhere around January. During the winter season there is decrease in solar irradiance so a wide difference in the relative energy difference between the measured and the stimulated daily energy yield can be observed.

When there is a 10% decrease in the tilt angle (here a 10% decrease was analyzed), the average relative daily difference decreased. The change in the relative daily difference and then the average value of the relative daily difference was calculated which is -0.9562082.

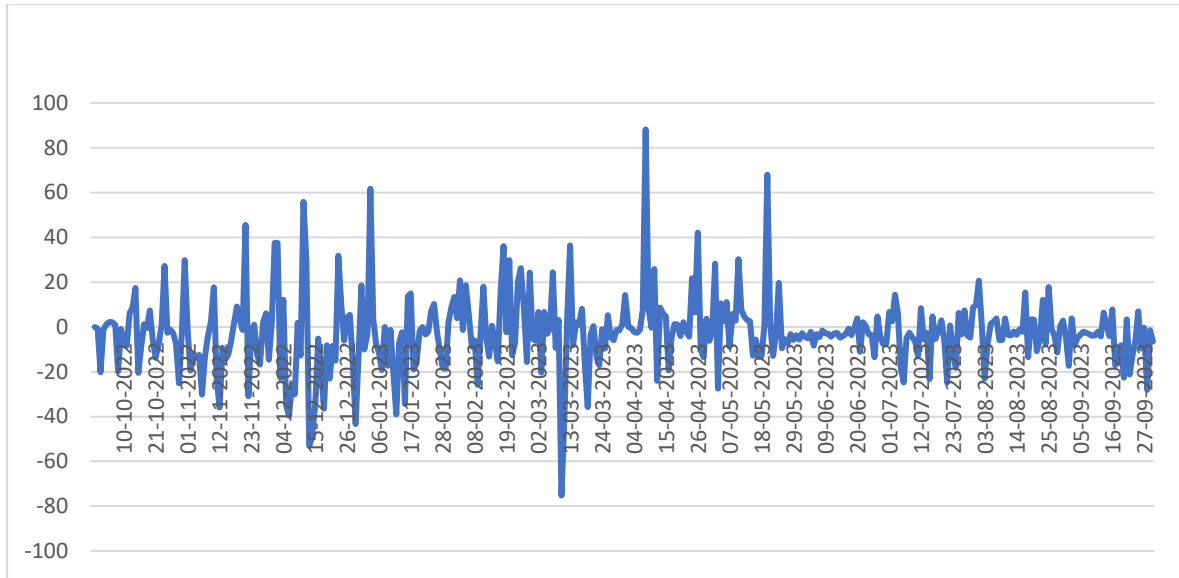


Fig 1: The relation between the relative daily difference between the measured and stimulated daily energy yield and the months of the year for one optimizer.

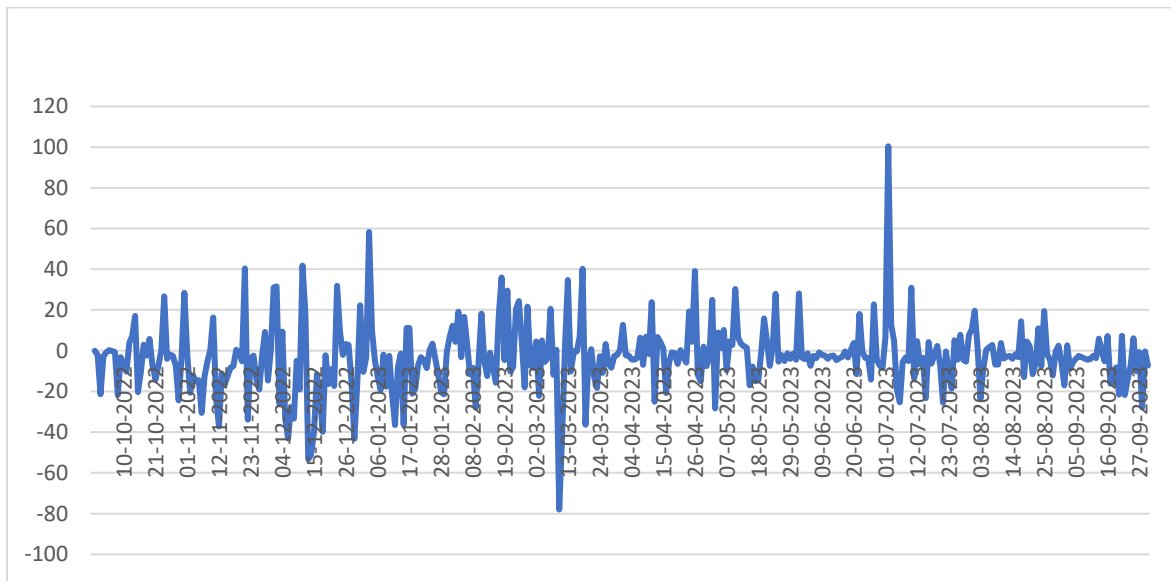


Fig 2: The relation between the relative daily difference between the measured and stimulated daily energy yield and the months of the year for another optimizer.

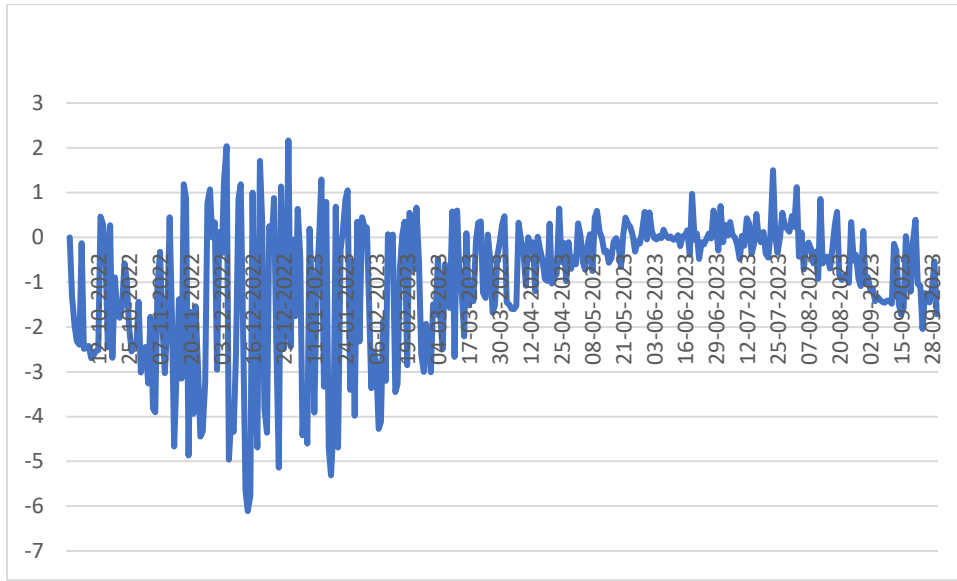


Fig 3: The relation between the relative daily difference between the measured and stimulated daily energy yield and the months of the year after making a change in the tilt angle by 10%.